Task 6 — Run 5

This document contains only Run 5's own implementations and test suites, exactly as executed. No files from any other run are included.

Test suite: amazon → Implementation: amazon

implementation_manual.py

```
def is_valid_password(password):
    if password is None or password == "":
        return ["Please fill the password"]
    errors = []
    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")
    if not any(char.islower() for char in password):
        errors.append("Password must contain a lowercase letter")
    if not any(char.isupper() for char in password):
        errors.append("Password must contain an uppercase letter")
    if not any(char.isdigit() or not char.isalpha() for char in password):
        errors.append("Password must contain a special character or a number")
    if not errors:
        return "The password is valid"
    return errors
```

selected_test.py

```
from unittest import TestCase
from implementation_manual import IsValidPassword

class TestIsValidPassword(TestCase):
    def test_missing_password(self):
        self.assertEqual(IsValidPassword(""), ["Please fill the password"])

    def test_password_too_short(self):
        self.assertEqual(IsValidPassword("short"), ["Length must be between 12 and 72
characters", "Password must contain an uppercase letter", "Password must contain a
special character or a number"])

    def test_password_too_long(self):
        self.assertEqual(IsValidPassword("a" * 73), ["Length must be between 12 and 72
characters"])

    def test_no_lowercase_letter(self):
        self.assertEqual(IsValidPassword("ABC123!"), ["Password must contain a
lowercase letter"])

    def test_no_uppercase_letter(self):
```

```
        self.assertEqual(IsValidPassword("abc123!"), ["Password must contain an
uppercase letter"])

    def test_no_special_character_or_number(self):
        self.assertEqual(IsValidPassword("abcABC"), ["Password must contain a special
character or a number"])

    def test_strong_password(self):
        self.assertEqual(IsValidPassword("aBcD123!@#"), "The password is valid")

    def test_multiple_errors(self):
        self.assertEqual(IsValidPassword("short"), ["Length must be between 12 and 72
characters", "Password must contain an uppercase letter", "Password must contain a
special character or a number"])
```

Test suite: amazon → Implementation: deepseek

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors

    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")

    has_lower = any(c.islower() for c in password)
    has_upper = any(c.isupper() for c in password)
    has_digit_or_special = any(not c.isalpha() for c in password)

    if not has_lower:
        errors.append("Password must contain a lowercase letter")
    if not has_upper:
        errors.append("Password must contain an uppercase letter")
    if not has_digit_or_special:
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"
    else:
        return errors
```

selected_test.py

```
from unittest import TestCase
from implementation_manual import IsValidPassword

class TestIsValidPassword(TestCase):
    def test_missing_password(self):
        self.assertEqual(IsValidPassword(""), ["Please fill the password"])

    def test_password_too_short(self):
        self.assertEqual(IsValidPassword("short"), ["Length must be between 12 and 72
characters", "Password must contain an uppercase letter", "Password must contain a
special character or a number"])

    def test_password_too_long(self):
        self.assertEqual(IsValidPassword("a" * 73), ["Length must be between 12 and 72
characters"])

    def test_no_lowercase_letter(self):
        self.assertEqual(IsValidPassword("ABC123!"), ["Password must contain a
lowercase letter"])

    def test_no_uppercase_letter(self):
        self.assertEqual(IsValidPassword("abc123!"), ["Password must contain an
```

```
uppercase letter"])

    def test_no_special_character_or_number(self):
        self.assertEqual(IsValidPassword("abcABC"), ["Password must contain a special
character or a number"])

    def test_strong_password(self):
        self.assertEqual(IsValidPassword("aBcD123!@#"), "The password is valid")

    def test_multiple_errors(self):
        self.assertEqual(IsValidPassword("short"), ["Length must be between 12 and 72
characters", "Password must contain an uppercase letter", "Password must contain a
special character or a number"])
```

Test suite: amazon → Implementation: gemma

implementation_manual.py

```
import re

def IsValidPassword(password):
    if password is None or password == "":
        return ["Please fill the password"]

    errors = []

    if not (12 <= len(password) <= 72):
        errors.append("Length must be between 12 and 72 characters")

    if not re.search(r"[a-z]", password):
        errors.append("Password must contain a lowercase letter")

    if not re.search(r"[A-Z]", password):
        errors.append("Password must contain an uppercase letter")

    if not re.search(r"[!?@#$%^&*()_+0-9\-=_~`\[\]{}|;':\",./<>?]", password) and not
any(char.isdigit() for char in password):
        # The requirement specifies: one digit or punctuation character ([!? @#$ %^&* _
0 -9])
        # Let's use a regex that covers digits and the specified special characters
        if not re.search(r"[!?@#$%^&* _0-9]", password):
            errors.append("Password must contain a special character or a number")

    # Re-evaluating the special character/number requirement based on the prompt's
specific list
    # Prompt: "one digit or punctuation character ([!? @#$ %^&* _ 0 -9])"
    # Let's refine the check to be exact to the prompt's logic

    # Resetting error list for the specific character check to ensure accuracy
    # We need to check lowercase, uppercase, and (digit OR special)

    # Re-implementing logic cleanly
    errors = []

    if not (12 <= len(password) <= 72):
        errors.append("Length must be between 12 and 72 characters")

    if not any(c.islower() for c in password):
        errors.append("Password must contain a lowercase letter")

    if not any(c.isupper() for c in password):
        errors.append("Password must contain an uppercase letter")

    # Check for digit or the specific special characters provided
    special_chars = "!?@#$%^&*_"
    has_digit = any(c.isdigit() for c in password)
    has_special = any(c in special_chars for c in password)
```

```
    if not (has_digit or has_special):
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"

    return errors
```

selected_test.py

```
from unittest import TestCase
from implementation_manual import IsValidPassword

class TestIsValidPassword(TestCase):
    def test_missing_password(self):
        self.assertEqual(IsValidPassword(""), ["Please fill the password"])

    def test_password_too_short(self):
        self.assertEqual(IsValidPassword("short"), ["Length must be between 12 and 72
characters", "Password must contain an uppercase letter", "Password must contain a
special character or a number"])

    def test_password_too_long(self):
        self.assertEqual(IsValidPassword("a" * 73), ["Length must be between 12 and 72
characters"])

    def test_no_lowercase_letter(self):
        self.assertEqual(IsValidPassword("ABC123!"), ["Password must contain a
lowercase letter"])

    def test_no_uppercase_letter(self):
        self.assertEqual(IsValidPassword("abc123!"), ["Password must contain an
uppercase letter"])

    def test_no_special_character_or_number(self):
        self.assertEqual(IsValidPassword("abcABC"), ["Password must contain a special
character or a number"])

    def test_strong_password(self):
        self.assertEqual(IsValidPassword("aBcD123!@#"), "The password is valid")

    def test_multiple_errors(self):
        self.assertEqual(IsValidPassword("short"), ["Length must be between 12 and 72
characters", "Password must contain an uppercase letter", "Password must contain a
special character or a number"])
```

Test suite: amazon → Implementation: manual

implementation_manual.py

```
def IsValidPassword(password):
    errors=[]
    if len(password)==0:
        return "Please fill the password"
    if len(password)<12 or len(password)>72:
        errors.append("Length must be between 12 and 72 characters")
    lowerCase=False
    upperCase=False
    num=False
    special=False
    for char in password:
        if(char.isdigit()):
            num=True
        if(char.islower()):
            lowerCase=True
        if(char.isupper()):
            upperCase=True
        if(not char.isalnum()):
            special=True
    if not lowerCase:
        errors.append("Password must contain a lowercase letter")
    if not upperCase:
        errors.append("Password must contain an uppercase letter")
    if not special or not num:
        errors.append("Password must contain a special character or a number")
    if len(errors)==0:
        return "The password is valid"
    else:
        return errors
```

selected_test.py

```
from unittest import TestCase
from implementation_manual import IsValidPassword

class TestIsValidPassword(TestCase):
    def test_missing_password(self):
        self.assertEqual(IsValidPassword(""), ["Please fill the password"])

    def test_password_too_short(self):
        self.assertEqual(IsValidPassword("short"), ["Length must be between 12 and 72
characters", "Password must contain an uppercase letter", "Password must contain a
special character or a number"])

    def test_password_too_long(self):
        self.assertEqual(IsValidPassword("a" * 73), ["Length must be between 12 and 72
characters"])
```

```
    def test_no_lowercase_letter(self):
        self.assertEqual(IsValidPassword("ABC123!"), ["Password must contain a
lowercase letter"])

    def test_no_uppercase_letter(self):
        self.assertEqual(IsValidPassword("abc123!"), ["Password must contain an
uppercase letter"])

    def test_no_special_character_or_number(self):
        self.assertEqual(IsValidPassword("abcABC"), ["Password must contain a special
character or a number"])

    def test_strong_password(self):
        self.assertEqual(IsValidPassword("aBcD123!@#"), "The password is valid")

    def test_multiple_errors(self):
        self.assertEqual(IsValidPassword("short"), ["Length must be between 12 and 72
characters", "Password must contain an uppercase letter", "Password must contain a
special character or a number"])
```

Test suite: amazon → Implementation: mistral

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors

    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")

    has_lower = any(c.islower() for c in password)
    if not has_lower:
        errors.append("Password must contain a lowercase letter")

    has_upper = any(c.isupper() for c in password)
    if not has_upper:
        errors.append("Password must contain an uppercase letter")

    has_special_or_digit = any(c.isdigit() or c in "!?@#$%^&*_-" for c in password)
    if not has_special_or_digit:
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"
    return errors
```

selected_test.py

```
from unittest import TestCase
from implementation_manual import IsValidPassword

class TestIsValidPassword(TestCase):
    def test_missing_password(self):
        self.assertEqual(IsValidPassword(""), ["Please fill the password"])

    def test_password_too_short(self):
        self.assertEqual(IsValidPassword("short"), ["Length must be between 12 and 72
characters", "Password must contain an uppercase letter", "Password must contain a
special character or a number"])

    def test_password_too_long(self):
        self.assertEqual(IsValidPassword("a" * 73), ["Length must be between 12 and 72
characters"])

    def test_no_lowercase_letter(self):
        self.assertEqual(IsValidPassword("ABC123!"), ["Password must contain a
lowercase letter"])

    def test_no_uppercase_letter(self):
        self.assertEqual(IsValidPassword("abc123!"), ["Password must contain an
```

```
uppercase letter"])

    def test_no_special_character_or_number(self):
        self.assertEqual(IsValidPassword("abcABC"), ["Password must contain a special
character or a number"])

    def test_strong_password(self):
        self.assertEqual(IsValidPassword("aBcD123!@#"), "The password is valid")

    def test_multiple_errors(self):
        self.assertEqual(IsValidPassword("short"), ["Length must be between 12 and 72
characters", "Password must contain an uppercase letter", "Password must contain a
special character or a number"])
```

Test suite: amazon → Implementation: openai

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors
    if not isinstance(password, str):
        return None
    length = len(password)
    if length < 12 or length > 72:
        errors.append("Length must be between 12 and 72 characters")
    if not any(c.islower() for c in password):
        errors.append("Password must contain a lowercase letter")
    if not any(c.isupper() for c in password):
        errors.append("Password must contain an uppercase letter")
    if not any(c.isdigit() or c in "!@#$%^&*_?" for c in password):
        errors.append("Password must contain a special character or a number")
    if errors:
        return errors
    return "The password is valid"
```

selected_test.py

```
from unittest import TestCase
from implementation_manual import IsValidPassword

class TestIsValidPassword(TestCase):
    def test_missing_password(self):
        self.assertEqual(IsValidPassword(""), ["Please fill the password"])

    def test_password_too_short(self):
        self.assertEqual(IsValidPassword("short"), ["Length must be between 12 and 72
characters", "Password must contain an uppercase letter", "Password must contain a
special character or a number"])

    def test_password_too_long(self):
        self.assertEqual(IsValidPassword("a" * 73), ["Length must be between 12 and 72
characters"])

    def test_no_lowercase_letter(self):
        self.assertEqual(IsValidPassword("ABC123!"), ["Password must contain a
lowercase letter"])

    def test_no_uppercase_letter(self):
        self.assertEqual(IsValidPassword("abc123!"), ["Password must contain an
uppercase letter"])

    def test_no_special_character_or_number(self):
        self.assertEqual(IsValidPassword("abcABC"), ["Password must contain a special
character or a number"])
```

```
    def test_strong_password(self):
        self.assertEqual(IsValidPassword("aBcD123!@#"), "The password is valid")

    def test_multiple_errors(self):
        self.assertEqual(IsValidPassword("short"), ["Length must be between 12 and 72
characters", "Password must contain an uppercase letter", "Password must contain a
special character or a number"])
```

Test suite: deepseek → Implementation: amazon

implementation_manual.py

```
def is_valid_password(password):
    if password is None or password == "":
        return ["Please fill the password"]
    errors = []
    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")
    if not any(char.islower() for char in password):
        errors.append("Password must contain a lowercase letter")
    if not any(char.isupper() for char in password):
        errors.append("Password must contain an uppercase letter")
    if not any(char.isdigit() or not char.isalpha() for char in password):
        errors.append("Password must contain a special character or a number")
    if not errors:
        return "The password is valid"
    return errors
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestPasswordValidator(unittest.TestCase):
    def test_missing_password(self):
        self.assertEqual(IsValidPassword(""), ["Please fill the password"])

    def test_password_too_short(self):
        self.assertEqual(IsValidPassword("Short1!"), ["Length must be between 12 and 72
characters"])

    def test_password_too_long(self):
        long_password = "a" * 73
        self.assertEqual(IsValidPassword(long_password), ["Length must be between 12
and 72 characters"])

    def test_password_missing_lowercase(self):
        password = "UPPERCASE123!"
        self.assertEqual(IsValidPassword(password), ["Password must contain a lowercase
letter"])

    def test_password_missing_uppercase(self):
        password = "lowercase123!"
        self.assertEqual(IsValidPassword(password), ["Password must contain an
uppercase letter"])

    def test_password_missing_special_or_number(self):
        password = "NoSpecialChars"
        self.assertEqual(IsValidPassword(password), ["Password must contain a special
character or a number"])
```

```
def test_multiple_errors(self):
    password = "short"
    expected_errors = [
        "Length must be between 12 and 72 characters",
        "Password must contain an uppercase letter",
        "Password must contain a special character or a number"
    ]
    self.assertEqual(IsValidPassword(password), expected_errors)

def test_valid_password(self):
    password = "ValidPassword123!"
    self.assertEqual(IsValidPassword(password), "The password is valid")

def test_valid_password_with_special_chars(self):
    password = "AnotherValid!@#123"
    self.assertEqual(IsValidPassword(password), "The password is valid")

def test_valid_password_max_length(self):
    password = "A" + "a" * 70 + "1!"
    self.assertEqual(IsValidPassword(password), "The password is valid")

def test_valid_password_min_length(self):
    password = "Aa1!Aa1!Aa1!A"
    self.assertEqual(IsValidPassword(password), "The password is valid")
```

Test suite: deepseek → Implementation: deepseek

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors

    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")

    has_lower = any(c.islower() for c in password)
    has_upper = any(c.isupper() for c in password)
    has_digit_or_special = any(not c.isalpha() for c in password)

    if not has_lower:
        errors.append("Password must contain a lowercase letter")
    if not has_upper:
        errors.append("Password must contain an uppercase letter")
    if not has_digit_or_special:
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"
    else:
        return errors
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestPasswordValidator(unittest.TestCase):
    def test_missing_password(self):
        self.assertEqual(IsValidPassword(""), ["Please fill the password"])

    def test_password_too_short(self):
        self.assertEqual(IsValidPassword("Short1!"), ["Length must be between 12 and 72
characters"])

    def test_password_too_long(self):
        long_password = "a" * 73
        self.assertEqual(IsValidPassword(long_password), ["Length must be between 12
and 72 characters"])

    def test_password_missing_lowercase(self):
        password = "UPPERCASE123!"
        self.assertEqual(IsValidPassword(password), ["Password must contain a lowercase
letter"])

    def test_password_missing_uppercase(self):
```

```
        password = "lowercase123!"
        self.assertEqual(IsValidPassword(password), ["Password must contain an
uppercase letter"])

    def test_password_missing_special_or_number(self):
        password = "NoSpecialChars"
        self.assertEqual(IsValidPassword(password), ["Password must contain a special
character or a number"])

    def test_multiple_errors(self):
        password = "short"
        expected_errors = [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ]
        self.assertEqual(IsValidPassword(password), expected_errors)

    def test_valid_password(self):
        password = "ValidPassword123!"
        self.assertEqual(IsValidPassword(password), "The password is valid")

    def test_valid_password_with_special_chars(self):
        password = "AnotherValid!@#123"
        self.assertEqual(IsValidPassword(password), "The password is valid")

    def test_valid_password_max_length(self):
        password = "A" + "a" * 70 + "1!"
        self.assertEqual(IsValidPassword(password), "The password is valid")

    def test_valid_password_min_length(self):
        password = "Aa1!Aa1!Aa1!A"
        self.assertEqual(IsValidPassword(password), "The password is valid")
```

Test suite: deepseek → Implementation: gemma

implementation_manual.py

```
import re

def IsValidPassword(password):
    if password is None or password == "":
        return ["Please fill the password"]

    errors = []

    if not (12 <= len(password) <= 72):
        errors.append("Length must be between 12 and 72 characters")

    if not re.search(r"[a-z]", password):
        errors.append("Password must contain a lowercase letter")

    if not re.search(r"[A-Z]", password):
        errors.append("Password must contain an uppercase letter")

    if not re.search(r"[!?@#$%^&*()_+0-9\-=_~`\[\]{}|;':\",./<>?]", password) and not
any(char.isdigit() for char in password):
        # The requirement specifies: one digit or punctuation character ([!? @#$ %^&* _
0 -9])
        # Let's use a regex that covers digits and the specified special characters
        if not re.search(r"[!?@#$%^&* _0-9]", password):
            errors.append("Password must contain a special character or a number")

    # Re-evaluating the special character/number requirement based on the prompt's
specific list
    # Prompt: "one digit or punctuation character ([!? @#$ %^&* _ 0 -9])"
    # Let's refine the check to be exact to the prompt's logic

    # Resetting error list for the specific character check to ensure accuracy
    # We need to check lowercase, uppercase, and (digit OR special)

    # Re-implementing logic cleanly
    errors = []

    if not (12 <= len(password) <= 72):
        errors.append("Length must be between 12 and 72 characters")

    if not any(c.islower() for c in password):
        errors.append("Password must contain a lowercase letter")

    if not any(c.isupper() for c in password):
        errors.append("Password must contain an uppercase letter")

    # Check for digit or the specific special characters provided
    special_chars = "!?@#$%^&*_"
    has_digit = any(c.isdigit() for c in password)
    has_special = any(c in special_chars for c in password)
```

```
    if not (has_digit or has_special):
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"

    return errors
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestPasswordValidator(unittest.TestCase):
    def test_missing_password(self):
        self.assertEqual(IsValidPassword(""), ["Please fill the password"])

    def test_password_too_short(self):
        self.assertEqual(IsValidPassword("Short1!"), ["Length must be between 12 and 72
characters"])

    def test_password_too_long(self):
        long_password = "a" * 73
        self.assertEqual(IsValidPassword(long_password), ["Length must be between 12
and 72 characters"])

    def test_password_missing_lowercase(self):
        password = "UPPERCASE123!"
        self.assertEqual(IsValidPassword(password), ["Password must contain a lowercase
letter"])

    def test_password_missing_uppercase(self):
        password = "lowercase123!"
        self.assertEqual(IsValidPassword(password), ["Password must contain an
uppercase letter"])

    def test_password_missing_special_or_number(self):
        password = "NoSpecialChars"
        self.assertEqual(IsValidPassword(password), ["Password must contain a special
character or a number"])

    def test_multiple_errors(self):
        password = "short"
        expected_errors = [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ]
        self.assertEqual(IsValidPassword(password), expected_errors)

    def test_valid_password(self):
        password = "ValidPassword123!"
```

```
    self.assertEqual(IsValidPassword(password), "The password is valid")

def test_valid_password_with_special_chars(self):
    password = "AnotherValid!@#123"
    self.assertEqual(IsValidPassword(password), "The password is valid")

def test_valid_password_max_length(self):
    password = "A" + "a" * 70 + "1!"
    self.assertEqual(IsValidPassword(password), "The password is valid")

def test_valid_password_min_length(self):
    password = "Aa1!Aa1!Aa1!A"
    self.assertEqual(IsValidPassword(password), "The password is valid")
```

Test suite: deepseek → Implementation: manual

implementation_manual.py

```
def IsValidPassword(password):
    errors=[]
    if len(password)==0:
        return "Please fill the password"
    if len(password)<12 or len(password)>72:
        errors.append("Length must be between 12 and 72 characters")
    lowerCase=False
    upperCase=False
    num=False
    special=False
    for char in password:
        if(char.isdigit()):
            num=True
        if(char.islower()):
            lowerCase=True
        if(char.isupper()):
            upperCase=True
        if(not char.isalnum()):
            special=True
    if not lowerCase:
        errors.append("Password must contain a lowercase letter")
    if not upperCase:
        errors.append("Password must contain an uppercase letter")
    if not special or not num:
        errors.append("Password must contain a special character or a number")
    if len(errors)==0:
        return "The password is valid"
    else:
        return errors
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestPasswordValidator(unittest.TestCase):
    def test_missing_password(self):
        self.assertEqual(IsValidPassword(""), ["Please fill the password"])

    def test_password_too_short(self):
        self.assertEqual(IsValidPassword("Short1!"), ["Length must be between 12 and 72
characters"])

    def test_password_too_long(self):
        long_password = "a" * 73
        self.assertEqual(IsValidPassword(long_password), ["Length must be between 12
and 72 characters"])
```

```
    def test_password_missing_lowercase(self):
        password = "UPPERCASE123!"
        self.assertEqual(IsValidPassword(password), ["Password must contain a lowercase
letter"])

    def test_password_missing_uppercase(self):
        password = "lowercase123!"
        self.assertEqual(IsValidPassword(password), ["Password must contain an
uppercase letter"])

    def test_password_missing_special_or_number(self):
        password = "NoSpecialChars"
        self.assertEqual(IsValidPassword(password), ["Password must contain a special
character or a number"])

    def test_multiple_errors(self):
        password = "short"
        expected_errors = [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ]
        self.assertEqual(IsValidPassword(password), expected_errors)

    def test_valid_password(self):
        password = "ValidPassword123!"
        self.assertEqual(IsValidPassword(password), "The password is valid")

    def test_valid_password_with_special_chars(self):
        password = "AnotherValid!@#123"
        self.assertEqual(IsValidPassword(password), "The password is valid")

    def test_valid_password_max_length(self):
        password = "A" + "a" * 70 + "1!"
        self.assertEqual(IsValidPassword(password), "The password is valid")

    def test_valid_password_min_length(self):
        password = "Aa1!Aa1!Aa1!A"
        self.assertEqual(IsValidPassword(password), "The password is valid")
```

Test suite: deepseek → Implementation: mistral

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors

    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")

    has_lower = any(c.islower() for c in password)
    if not has_lower:
        errors.append("Password must contain a lowercase letter")

    has_upper = any(c.isupper() for c in password)
    if not has_upper:
        errors.append("Password must contain an uppercase letter")

    has_special_or_digit = any(c.isdigit() or c in "!?@#$%^&*_-" for c in password)
    if not has_special_or_digit:
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"
    return errors
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestPasswordValidator(unittest.TestCase):
    def test_missing_password(self):
        self.assertEqual(IsValidPassword(""), ["Please fill the password"])

    def test_password_too_short(self):
        self.assertEqual(IsValidPassword("Short1!"), ["Length must be between 12 and 72
characters"])

    def test_password_too_long(self):
        long_password = "a" * 73
        self.assertEqual(IsValidPassword(long_password), ["Length must be between 12
and 72 characters"])

    def test_password_missing_lowercase(self):
        password = "UPPERCASE123!"
        self.assertEqual(IsValidPassword(password), ["Password must contain a lowercase
letter"])

    def test_password_missing_uppercase(self):
```

```
        password = "lowercase123!"
        self.assertEqual(IsValidPassword(password), ["Password must contain an
uppercase letter"])

    def test_password_missing_special_or_number(self):
        password = "NoSpecialChars"
        self.assertEqual(IsValidPassword(password), ["Password must contain a special
character or a number"])

    def test_multiple_errors(self):
        password = "short"
        expected_errors = [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ]
        self.assertEqual(IsValidPassword(password), expected_errors)

    def test_valid_password(self):
        password = "ValidPassword123!"
        self.assertEqual(IsValidPassword(password), "The password is valid")

    def test_valid_password_with_special_chars(self):
        password = "AnotherValid!@#123"
        self.assertEqual(IsValidPassword(password), "The password is valid")

    def test_valid_password_max_length(self):
        password = "A" + "a" * 70 + "1!"
        self.assertEqual(IsValidPassword(password), "The password is valid")

    def test_valid_password_min_length(self):
        password = "Aa1!Aa1!Aa1!A"
        self.assertEqual(IsValidPassword(password), "The password is valid")
```

Test suite: deepseek → Implementation: openai

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors
    if not isinstance(password, str):
        return None
    length = len(password)
    if length < 12 or length > 72:
        errors.append("Length must be between 12 and 72 characters")
    if not any(c.islower() for c in password):
        errors.append("Password must contain a lowercase letter")
    if not any(c.isupper() for c in password):
        errors.append("Password must contain an uppercase letter")
    if not any(c.isdigit() or c in "!@#$%^&*_?" for c in password):
        errors.append("Password must contain a special character or a number")
    if errors:
        return errors
    return "The password is valid"
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestPasswordValidator(unittest.TestCase):
    def test_missing_password(self):
        self.assertEqual(IsValidPassword(""), ["Please fill the password"])

    def test_password_too_short(self):
        self.assertEqual(IsValidPassword("Short1!"), ["Length must be between 12 and 72
characters"])

    def test_password_too_long(self):
        long_password = "a" * 73
        self.assertEqual(IsValidPassword(long_password), ["Length must be between 12
and 72 characters"])

    def test_password_missing_lowercase(self):
        password = "UPPERCASE123!"
        self.assertEqual(IsValidPassword(password), ["Password must contain a lowercase
letter"])

    def test_password_missing_uppercase(self):
        password = "lowercase123!"
        self.assertEqual(IsValidPassword(password), ["Password must contain an
uppercase letter"])

    def test_password_missing_special_or_number(self):
```

```
        password = "NoSpecialChars"
        self.assertEqual(IsValidPassword(password), ["Password must contain a special
character or a number"])

    def test_multiple_errors(self):
        password = "short"
        expected_errors = [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ]
        self.assertEqual(IsValidPassword(password), expected_errors)

    def test_valid_password(self):
        password = "ValidPassword123!"
        self.assertEqual(IsValidPassword(password), "The password is valid")

    def test_valid_password_with_special_chars(self):
        password = "AnotherValid!@#123"
        self.assertEqual(IsValidPassword(password), "The password is valid")

    def test_valid_password_max_length(self):
        password = "A" + "a" * 70 + "1!"
        self.assertEqual(IsValidPassword(password), "The password is valid")

    def test_valid_password_min_length(self):
        password = "Aa1!Aa1!Aa1!A"
        self.assertEqual(IsValidPassword(password), "The password is valid")
```

Test suite: gemma → Implementation: amazon

implementation_manual.py

```
def is_valid_password(password):
    if password is None or password == "":
        return ["Please fill the password"]
    errors = []
    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")
    if not any(char.islower() for char in password):
        errors.append("Password must contain a lowercase letter")
    if not any(char.isupper() for char in password):
        errors.append("Password must contain an uppercase letter")
    if not any(char.isdigit() or not char.isalpha() for char in password):
        errors.append("Password must contain a special character or a number")
    if not errors:
        return "The password is valid"
    return errors
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestIsValidPassword(unittest.TestCase):

    def test_valid_password_returns_success_message(self):
        # Arrange
        password = "StrongPassword123!"
        expected = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_password_returns_fill_error(self):
        # Arrange
        password = ""
        expected = ["Please fill the password"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_none_password_returns_fill_error(self):
        # Arrange
        password = None
```

```
        expected = ["Please fill the password"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_password_too_short_returns_length_error(self):
        # Arrange
        password = "Ab1!"
        expected = ["Length must be between 12 and 72 characters"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_password_too_long_returns_length_error(self):
        # Arrange
        password = "A" * 73 + "a1!"
        expected = ["Length must be between 12 and 72 characters"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_lowercase_returns_lowercase_error(self):
        # Arrange
        password = "STRONGPASS123!"
        expected = ["Password must contain a lowercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_uppercase_returns_uppercase_error(self):
        # Arrange
        password = "strongpass123!"
        expected = ["Password must contain an uppercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_special_or_number_returns_special_error(self):
```

```
        # Arrange
        password = "StrongPassword"
        expected = ["Password must contain a special character or a number"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_multiple_validation_failures(self):
        # Arrange
        # Short, no uppercase, no special/number
        password = "abc"
        expected = [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ]

        # Act
        result = IsValidPassword(password)

        # Assert
        # We check if all expected errors are present since order might vary depending
on implementation
        for error in expected:
            self.assertIn(error, result)
        self.assertEqual(len(result), len(expected))

    def test_boundary_length_12_is_valid(self):
        # Arrange
        password = "Abcdefghij1!"
        expected = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_boundary_length_72_is_valid(self):
        # Arrange
        password = "A" * 70 + "a1!"
        expected = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)
```

```
if __name__ == '__main__':
    unittest.main()
```

Test suite: gemma → Implementation: deepseek

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors

    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")

    has_lower = any(c.islower() for c in password)
    has_upper = any(c.isupper() for c in password)
    has_digit_or_special = any(not c.isalpha() for c in password)

    if not has_lower:
        errors.append("Password must contain a lowercase letter")
    if not has_upper:
        errors.append("Password must contain an uppercase letter")
    if not has_digit_or_special:
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"
    else:
        return errors
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestIsValidPassword(unittest.TestCase):

    def test_valid_password_returns_success_message(self):
        # Arrange
        password = "StrongPassword123!"
        expected = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_password_returns_fill_error(self):
        # Arrange
        password = ""
        expected = ["Please fill the password"]

        # Act
```

```
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_none_password_returns_fill_error(self):
        # Arrange
        password = None
        expected = ["Please fill the password"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_password_too_short_returns_length_error(self):
        # Arrange
        password = "Ab1!"
        expected = ["Length must be between 12 and 72 characters"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_password_too_long_returns_length_error(self):
        # Arrange
        password = "A" * 73 + "a1!"
        expected = ["Length must be between 12 and 72 characters"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_lowercase_returns_lowercase_error(self):
        # Arrange
        password = "STRONGPASS123!"
        expected = ["Password must contain a lowercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_uppercase_returns_uppercase_error(self):
        # Arrange
        password = "strongpass123!"
        expected = ["Password must contain an uppercase letter"]
```

```
        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_special_or_number_returns_special_error(self):
        # Arrange
        password = "StrongPassword"
        expected = ["Password must contain a special character or a number"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_multiple_validation_failures(self):
        # Arrange
        # Short, no uppercase, no special/number
        password = "abc"
        expected = [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ]

        # Act
        result = IsValidPassword(password)

        # Assert
        # We check if all expected errors are present since order might vary depending
on implementation
        for error in expected:
            self.assertIn(error, result)
        self.assertEqual(len(result), len(expected))

    def test_boundary_length_12_is_valid(self):
        # Arrange
        password = "Abcdefghij1!"
        expected = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_boundary_length_72_is_valid(self):
        # Arrange
        password = "A" * 70 + "a1!"
        expected = "The password is valid"
```

```
        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

if __name__ == '__main__':
    unittest.main()
```

Test suite: gemma → Implementation: gemma

implementation_manual.py

```
import re

def IsValidPassword(password):
    if password is None or password == "":
        return ["Please fill the password"]

    errors = []

    if not (12 <= len(password) <= 72):
        errors.append("Length must be between 12 and 72 characters")

    if not re.search(r"[a-z]", password):
        errors.append("Password must contain a lowercase letter")

    if not re.search(r"[A-Z]", password):
        errors.append("Password must contain an uppercase letter")

    if not re.search(r"[!?@#$%^&*()_+0-9\-=_~`\[\]{}|;':\",./<>?]", password) and not
any(char.isdigit() for char in password):
        # The requirement specifies: one digit or punctuation character ([!? @#$ %^&* _
0 -9])
        # Let's use a regex that covers digits and the specified special characters
        if not re.search(r"[!?@#$%^&* _0-9]", password):
            errors.append("Password must contain a special character or a number")

    # Re-evaluating the special character/number requirement based on the prompt's
specific list
    # Prompt: "one digit or punctuation character ([!? @#$ %^&* _ 0 -9])"
    # Let's refine the check to be exact to the prompt's logic

    # Resetting error list for the specific character check to ensure accuracy
    # We need to check lowercase, uppercase, and (digit OR special)

    # Re-implementing logic cleanly
    errors = []

    if not (12 <= len(password) <= 72):
        errors.append("Length must be between 12 and 72 characters")

    if not any(c.islower() for c in password):
        errors.append("Password must contain a lowercase letter")

    if not any(c.isupper() for c in password):
        errors.append("Password must contain an uppercase letter")

    # Check for digit or the specific special characters provided
    special_chars = "!?@#$%^&*_"
    has_digit = any(c.isdigit() for c in password)
    has_special = any(c in special_chars for c in password)
```

```
    if not (has_digit or has_special):
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"

    return errors
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestIsValidPassword(unittest.TestCase):

    def test_valid_password_returns_success_message(self):
        # Arrange
        password = "StrongPassword123!"
        expected = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_password_returns_fill_error(self):
        # Arrange
        password = ""
        expected = ["Please fill the password"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_none_password_returns_fill_error(self):
        # Arrange
        password = None
        expected = ["Please fill the password"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_password_too_short_returns_length_error(self):
        # Arrange
        password = "Ab1!"
        expected = ["Length must be between 12 and 72 characters"]
```

```
        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_password_too_long_returns_length_error(self):
        # Arrange
        password = "A" * 73 + "a1!"
        expected = ["Length must be between 12 and 72 characters"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_lowercase_returns_lowercase_error(self):
        # Arrange
        password = "STRONGPASS123!"
        expected = ["Password must contain a lowercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_uppercase_returns_uppercase_error(self):
        # Arrange
        password = "strongpass123!"
        expected = ["Password must contain an uppercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_special_or_number_returns_special_error(self):
        # Arrange
        password = "StrongPassword"
        expected = ["Password must contain a special character or a number"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_multiple_validation_failures(self):
        # Arrange
```

```
        # Short, no uppercase, no special/number
        password = "abc"
        expected = [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ]

        # Act
        result = IsValidPassword(password)

        # Assert
        # We check if all expected errors are present since order might vary depending
on implementation
        for error in expected:
            self.assertIn(error, result)
        self.assertEqual(len(result), len(expected))

    def test_boundary_length_12_is_valid(self):
        # Arrange
        password = "Abcdefghij1!"
        expected = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_boundary_length_72_is_valid(self):
        # Arrange
        password = "A" * 70 + "a1!"
        expected = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

if __name__ == '__main__':
    unittest.main()
```

Test suite: gemma → Implementation: manual

implementation_manual.py

```
def IsValidPassword(password):
    errors=[]
    if len(password)==0:
        return "Please fill the password"
    if len(password)<12 or len(password)>72:
        errors.append("Length must be between 12 and 72 characters")
    lowerCase=False
    upperCase=False
    num=False
    special=False
    for char in password:
        if(char.isdigit()):
            num=True
        if(char.islower()):
            lowerCase=True
        if(char.isupper()):
            upperCase=True
        if(not char.isalnum()):
            special=True
    if not lowerCase:
        errors.append("Password must contain a lowercase letter")
    if not upperCase:
        errors.append("Password must contain an uppercase letter")
    if not special or not num:
        errors.append("Password must contain a special character or a number")
    if len(errors)==0:
        return "The password is valid"
    else:
        return errors
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestIsValidPassword(unittest.TestCase):

    def test_valid_password_returns_success_message(self):
        # Arrange
        password = "StrongPassword123!"
        expected = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)
```

```
def test_missing_password_returns_fill_error(self):
    # Arrange
    password = ""
    expected = ["Please fill the password"]

    # Act
    result = IsValidPassword(password)

    # Assert
    self.assertEqual(result, expected)

def test_none_password_returns_fill_error(self):
    # Arrange
    password = None
    expected = ["Please fill the password"]

    # Act
    result = IsValidPassword(password)

    # Assert
    self.assertEqual(result, expected)

def test_password_too_short_returns_length_error(self):
    # Arrange
    password = "Ab1!"
    expected = ["Length must be between 12 and 72 characters"]

    # Act
    result = IsValidPassword(password)

    # Assert
    self.assertEqual(result, expected)

def test_password_too_long_returns_length_error(self):
    # Arrange
    password = "A" * 73 + "a1!"
    expected = ["Length must be between 12 and 72 characters"]

    # Act
    result = IsValidPassword(password)

    # Assert
    self.assertEqual(result, expected)

def test_missing_lowercase_returns_lowercase_error(self):
    # Arrange
    password = "STRONGPASS123!"
    expected = ["Password must contain a lowercase letter"]

    # Act
    result = IsValidPassword(password)
```

```
        # Assert
        self.assertEqual(result, expected)

    def test_missing_uppercase_returns_uppercase_error(self):
        # Arrange
        password = "strongpass123!"
        expected = ["Password must contain an uppercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_special_or_number_returns_special_error(self):
        # Arrange
        password = "StrongPassword"
        expected = ["Password must contain a special character or a number"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_multiple_validation_failures(self):
        # Arrange
        # Short, no uppercase, no special/number
        password = "abc"
        expected = [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ]

        # Act
        result = IsValidPassword(password)

        # Assert
        # We check if all expected errors are present since order might vary depending
on implementation
        for error in expected:
            self.assertIn(error, result)
        self.assertEqual(len(result), len(expected))

    def test_boundary_length_12_is_valid(self):
        # Arrange
        password = "Abcdefghij1!"
        expected = "The password is valid"

        # Act
        result = IsValidPassword(password)
```

```
        # Assert
        self.assertEqual(result, expected)

    def test_boundary_length_72_is_valid(self):
        # Arrange
        password = "A" * 70 + "a1!"
        expected = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

if __name__ == '__main__':
    unittest.main()
```

Test suite: gemma → Implementation: mistral

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors

    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")

    has_lower = any(c.islower() for c in password)
    if not has_lower:
        errors.append("Password must contain a lowercase letter")

    has_upper = any(c.isupper() for c in password)
    if not has_upper:
        errors.append("Password must contain an uppercase letter")

    has_special_or_digit = any(c.isdigit() or c in "!?@#$%^&*_-" for c in password)
    if not has_special_or_digit:
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"
    return errors
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestIsValidPassword(unittest.TestCase):

    def test_valid_password_returns_success_message(self):
        # Arrange
        password = "StrongPassword123!"
        expected = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_password_returns_fill_error(self):
        # Arrange
        password = ""
        expected = ["Please fill the password"]

        # Act
```

```
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_none_password_returns_fill_error(self):
        # Arrange
        password = None
        expected = ["Please fill the password"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_password_too_short_returns_length_error(self):
        # Arrange
        password = "Ab1!"
        expected = ["Length must be between 12 and 72 characters"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_password_too_long_returns_length_error(self):
        # Arrange
        password = "A" * 73 + "a1!"
        expected = ["Length must be between 12 and 72 characters"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_lowercase_returns_lowercase_error(self):
        # Arrange
        password = "STRONGPASS123!"
        expected = ["Password must contain a lowercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_uppercase_returns_uppercase_error(self):
        # Arrange
        password = "strongpass123!"
        expected = ["Password must contain an uppercase letter"]
```

```
        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_special_or_number_returns_special_error(self):
        # Arrange
        password = "StrongPassword"
        expected = ["Password must contain a special character or a number"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_multiple_validation_failures(self):
        # Arrange
        # Short, no uppercase, no special/number
        password = "abc"
        expected = [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ]

        # Act
        result = IsValidPassword(password)

        # Assert
        # We check if all expected errors are present since order might vary depending
on implementation
        for error in expected:
            self.assertIn(error, result)
        self.assertEqual(len(result), len(expected))

    def test_boundary_length_12_is_valid(self):
        # Arrange
        password = "Abcdefghij1!"
        expected = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_boundary_length_72_is_valid(self):
        # Arrange
        password = "A" * 70 + "a1!"
        expected = "The password is valid"
```

```
        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

if __name__ == '__main__':
    unittest.main()
```

Test suite: gemma → Implementation: openai

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors
    if not isinstance(password, str):
        return None
    length = len(password)
    if length < 12 or length > 72:
        errors.append("Length must be between 12 and 72 characters")
    if not any(c.islower() for c in password):
        errors.append("Password must contain a lowercase letter")
    if not any(c.isupper() for c in password):
        errors.append("Password must contain an uppercase letter")
    if not any(c.isdigit() or c in "!@#$%^&*_?" for c in password):
        errors.append("Password must contain a special character or a number")
    if errors:
        return errors
    return "The password is valid"
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestIsValidPassword(unittest.TestCase):

    def test_valid_password_returns_success_message(self):
        # Arrange
        password = "StrongPassword123!"
        expected = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_missing_password_returns_fill_error(self):
        # Arrange
        password = ""
        expected = ["Please fill the password"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)
```

```
def test_none_password_returns_fill_error(self):
    # Arrange
    password = None
    expected = ["Please fill the password"]

    # Act
    result = IsValidPassword(password)

    # Assert
    self.assertEqual(result, expected)

def test_password_too_short_returns_length_error(self):
    # Arrange
    password = "Ab1!"
    expected = ["Length must be between 12 and 72 characters"]

    # Act
    result = IsValidPassword(password)

    # Assert
    self.assertEqual(result, expected)

def test_password_too_long_returns_length_error(self):
    # Arrange
    password = "A" * 73 + "a1!"
    expected = ["Length must be between 12 and 72 characters"]

    # Act
    result = IsValidPassword(password)

    # Assert
    self.assertEqual(result, expected)

def test_missing_lowercase_returns_lowercase_error(self):
    # Arrange
    password = "STRONGPASS123!"
    expected = ["Password must contain a lowercase letter"]

    # Act
    result = IsValidPassword(password)

    # Assert
    self.assertEqual(result, expected)

def test_missing_uppercase_returns_uppercase_error(self):
    # Arrange
    password = "strongpass123!"
    expected = ["Password must contain an uppercase letter"]

    # Act
    result = IsValidPassword(password)

    # Assert
```

```
        self.assertEqual(result, expected)

    def test_missing_special_or_number_returns_special_error(self):
        # Arrange
        password = "StrongPassword"
        expected = ["Password must contain a special character or a number"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_multiple_validation_failures(self):
        # Arrange
        # Short, no uppercase, no special/number
        password = "abc"
        expected = [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ]

        # Act
        result = IsValidPassword(password)

        # Assert
        # We check if all expected errors are present since order might vary depending
on implementation
        for error in expected:
            self.assertIn(error, result)
        self.assertEqual(len(result), len(expected))

    def test_boundary_length_12_is_valid(self):
        # Arrange
        password = "Abcdefghij1!"
        expected = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected)

    def test_boundary_length_72_is_valid(self):
        # Arrange
        password = "A" * 70 + "a1!"
        expected = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
```

```
        self.assertEqual(result, expected)

if __name__ == '__main__':
    unittest.main()
```

Test suite: manual → Implementation: amazon

implementation_manual.py

```
def is_valid_password(password):
    if password is None or password == "":
        return ["Please fill the password"]
    errors = []
    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")
    if not any(char.islower() for char in password):
        errors.append("Password must contain a lowercase letter")
    if not any(char.isupper() for char in password):
        errors.append("Password must contain an uppercase letter")
    if not any(char.isdigit() or not char.isalpha() for char in password):
        errors.append("Password must contain a special character or a number")
    if not errors:
        return "The password is valid"
    return errors
```

selected_test.py

```
# Tests for task6_validate_password
import unittest
from implementation_manual import IsValidPassword

class Test_password_valid(unittest.TestCase):
    def test_no_password(self):
        self.assertEqual(IsValidPassword(""),"Please fill the password")

    def test_no_lowercase(self):
        self.assertEqual(IsValidPassword("ABCDEFGHIJK*1"),['Password must contain a
lowercase letter'])

    def test_no_upprcase(self):
        self.assertEqual(IsValidPassword("abcdefghijk*1"),['Password must contain an
uppercase letter'])

    def test_no_symbols(self):
        self.assertEqual(IsValidPassword("Abcdefghijkl1"),['Password must contain a
special character or a number'])

    def test_lower_upper_no_symbol(self):
        self.assertEqual(IsValidPassword("Abcdefghijkl1"),['Password must contain a
special character or a number'])

    def test_nolower_upper_symbol(self):
        self.assertEqual(IsValidPassword("ABCDEFGHIJK*1"),['Password must contain a
lowercase letter'])

    def test_noupper_lower_symbol(self):
        self.assertEqual(IsValidPassword("abcdefghijk*1"),['Password must contain an
```

```
uppercase letter'])

    def test_diff_language(self):
        self.assertEqual(IsValidPassword("ಅಆಇಈಉಊಋಎಏ12ಐ"),"The password is valid")

    def test_short_len_pwd(self):
        self.assertEqual(IsValidPassword("Shean123*"),['Length must be between 12 and
72 characters'])

    def test_long_len_pwd(self):
        self.assertEqual(IsValidPassword("Shean123*" + "a"*70),['Length must be between
12 and 72 characters'])

    def test_working_pwd(self):
        self.assertEqual(IsValidPassword("Shean12354*ab"),"The password is valid")

    def test_mix_languages(self):
        self.assertEqual(IsValidPassword("ಅಆಇಈ1231oremipsumZZ*"),"The password is
valid")

    def test_only_symbols(self):
        self.assertEqual(IsValidPassword("!@#$%^&*()*!@"),['Password must contain a
lowercase letter','Password must contain an uppercase letter','Password must contain a
special character or a number'])

    def test_spaces_in_pwd(self):
        self.assertEqual(IsValidPassword("Shean 12354*ab"),"The password is valid")

if __name__ == "__main__":
    unittest.main()
```

Test suite: manual → Implementation: deepseek

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors

    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")

    has_lower = any(c.islower() for c in password)
    has_upper = any(c.isupper() for c in password)
    has_digit_or_special = any(not c.isalpha() for c in password)

    if not has_lower:
        errors.append("Password must contain a lowercase letter")
    if not has_upper:
        errors.append("Password must contain an uppercase letter")
    if not has_digit_or_special:
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"
    else:
        return errors
```

selected_test.py

```
# Tests for task6_validate_password
import unittest
from implementation_manual import IsValidPassword

class Test_password_valid(unittest.TestCase):
    def test_no_password(self):
        self.assertEqual(IsValidPassword(""),"Please fill the password")

    def test_no_lowercase(self):
        self.assertEqual(IsValidPassword("ABCDEFGHIJK*1"),['Password must contain a
lowercase letter'])

    def test_no_upprcase(self):
        self.assertEqual(IsValidPassword("abcdefghijk*1"),['Password must contain an
uppercase letter'])

    def test_no_symbols(self):
        self.assertEqual(IsValidPassword("Abcdefghijkl1"),['Password must contain a
special character or a number'])

    def test_lower_upper_no_symbol(self):
        self.assertEqual(IsValidPassword("Abcdefghijkl1"),['Password must contain a
```

```
special character or a number'])

    def test_nolower_upper_symbol(self):
        self.assertEqual(IsValidPassword("ABCDEFGHIJK*1"),['Password must contain a
lowercase letter'])

    def test_noupper_lower_symbol(self):
        self.assertEqual(IsValidPassword("abcdefghijk*1"),['Password must contain an
uppercase letter'])

    def test_diff_language(self):
        self.assertEqual(IsValidPassword("ಅಆಇಈಉಊಋ1Aa"),"The password is valid")

    def test_short_len_pwd(self):
        self.assertEqual(IsValidPassword("Shean123*"),['Length must be between 12 and
72 characters'])

    def test_long_len_pwd(self):
        self.assertEqual(IsValidPassword("Shean123*" + "a"*70),['Length must be between
12 and 72 characters'])

    def test_working_pwd(self):
        self.assertEqual(IsValidPassword("Shean12354*ab"),"The password is valid")

    def test_mix_languages(self):
        self.assertEqual(IsValidPassword("ಅಆಇಈ1231oremipsumZZ*"),"The password is
valid")

    def test_only_symbols(self):
        self.assertEqual(IsValidPassword("!@#$%^&*()*!@"),['Password must contain a
lowercase letter','Password must contain an uppercase letter','Password must contain a
special character or a number'])

    def test_spaces_in_pwd(self):
        self.assertEqual(IsValidPassword("Shean 12354*ab"),"The password is valid")

if __name__ == "__main__":
    unittest.main()
```

Test suite: manual → Implementation: gemma

implementation_manual.py

```
import re

def IsValidPassword(password):
    if password is None or password == "":
        return ["Please fill the password"]

    errors = []

    if not (12 <= len(password) <= 72):
        errors.append("Length must be between 12 and 72 characters")

    if not re.search(r"[a-z]", password):
        errors.append("Password must contain a lowercase letter")

    if not re.search(r"[A-Z]", password):
        errors.append("Password must contain an uppercase letter")

    if not re.search(r"[!?@#$%^&*()_+0-9\-=_~`\[\]{}|;':\",./<>?]", password) and not
any(char.isdigit() for char in password):
        # The requirement specifies: one digit or punctuation character ([!? @#$ %^&* _
0 -9])
        # Let's use a regex that covers digits and the specified special characters
        if not re.search(r"[!?@#$%^&* _0-9]", password):
            errors.append("Password must contain a special character or a number")

    # Re-evaluating the special character/number requirement based on the prompt's
specific list
    # Prompt: "one digit or punctuation character ([!? @#$ %^&* _ 0 -9])"
    # Let's refine the check to be exact to the prompt's logic

    # Resetting error list for the specific character check to ensure accuracy
    # We need to check lowercase, uppercase, and (digit OR special)

    # Re-implementing logic cleanly
    errors = []

    if not (12 <= len(password) <= 72):
        errors.append("Length must be between 12 and 72 characters")

    if not any(c.islower() for c in password):
        errors.append("Password must contain a lowercase letter")

    if not any(c.isupper() for c in password):
        errors.append("Password must contain an uppercase letter")

    # Check for digit or the specific special characters provided
    special_chars = "!?@#$%^&*_"
    has_digit = any(c.isdigit() for c in password)
    has_special = any(c in special_chars for c in password)
```

```
    if not (has_digit or has_special):
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"

    return errors
```

selected_test.py

```
# Tests for task6_validate_password
import unittest
from implementation_manual import IsValidPassword

class Test_password_valid(unittest.TestCase):
    def test_no_password(self):
        self.assertEqual(IsValidPassword(""),"Please fill the password")

    def test_no_lowercase(self):
        self.assertEqual(IsValidPassword("ABCDEFGHIJK*1"),['Password must contain a
lowercase letter'])

    def test_no_upprcase(self):
        self.assertEqual(IsValidPassword("abcdefghijk*1"),['Password must contain an
uppercase letter'])

    def test_no_symbols(self):
        self.assertEqual(IsValidPassword("Abcdefghijkl1"),['Password must contain a
special character or a number'])

    def test_lower_upper_no_symbol(self):
        self.assertEqual(IsValidPassword("Abcdefghijkl1"),['Password must contain a
special character or a number'])

    def test_nolower_upper_symbol(self):
        self.assertEqual(IsValidPassword("ABCDEFGHIJK*1"),['Password must contain a
lowercase letter'])

    def test_noupper_lower_symbol(self):
        self.assertEqual(IsValidPassword("abcdefghijk*1"),['Password must contain an
uppercase letter'])

    def test_diff_language(self):
        self.assertEqual(IsValidPassword("[illegible]The password is valid")

    def test_short_len_pwd(self):
        self.assertEqual(IsValidPassword("Shean123*"),['Length must be between 12 and
72 characters'])

    def test_long_len_pwd(self):
        self.assertEqual(IsValidPassword("Shean123*" + "a"*70),['Length must be between
12 and 72 characters'])
```

```
    def test_working_pwd(self):
        self.assertEqual(IsValidPassword("Shean12354*ab"),"The password is valid")

    def test_mix_languages(self):
        self.assertEqual(IsValidPassword("ಅಆಇಈ1234loremipsumZZ*"),"The password is
valid")

    def test_only_symbols(self):
        self.assertEqual(IsValidPassword("!@#$%^&*()*!@"),['Password must contain a
lowercase letter','Password must contain an uppercase letter','Password must contain a
special character or a number'])

    def test_spaces_in_pwd(self):
        self.assertEqual(IsValidPassword("Shean 12354*ab"),"The password is valid")

if __name__ == "__main__":
    unittest.main()
```

Test suite: manual → Implementation: manual

implementation_manual.py

```
def IsValidPassword(password):
    errors=[]
    if len(password)==0:
        return "Please fill the password"
    if len(password)<12 or len(password)>72:
        errors.append("Length must be between 12 and 72 characters")
    lowerCase=False
    upperCase=False
    num=False
    special=False
    for char in password:
        if(char.isdigit()):
            num=True
        if(char.islower()):
            lowerCase=True
        if(char.isupper()):
            upperCase=True
        if(not char.isalnum()):
            special=True
    if not lowerCase:
        errors.append("Password must contain a lowercase letter")
    if not upperCase:
        errors.append("Password must contain an uppercase letter")
    if not special or not num:
        errors.append("Password must contain a special character or a number")
    if len(errors)==0:
        return "The password is valid"
    else:
        return errors
```

selected_test.py

```
# Tests for task6_validate_password
import unittest
from implementation_manual import IsValidPassword

class Test_password_valid(unittest.TestCase):
    def test_no_password(self):
        self.assertEqual(IsValidPassword(""),"Please fill the password")

    def test_no_lowercase(self):
        self.assertEqual(IsValidPassword("ABCDEFGHIJK*1"),['Password must contain a
lowercase letter'])

    def test_no_upprcase(self):
        self.assertEqual(IsValidPassword("abcdefghijk*1"),['Password must contain an
uppercase letter'])
```

```
    def test_no_symbols(self):
        self.assertEqual(IsValidPassword("Abcdefghijkl1"),['Password must contain a
special character or a number'])

    def test_lower_upper_no_symbol(self):
        self.assertEqual(IsValidPassword("Abcdefghijkl1"),['Password must contain a
special character or a number'])

    def test_nolower_upper_symbol(self):
        self.assertEqual(IsValidPassword("ABCDEFGHIJK*1"),['Password must contain a
lowercase letter'])

    def test_noupper_lower_symbol(self):
        self.assertEqual(IsValidPassword("abcdefghijk*1"),['Password must contain an
uppercase letter'])

    def test_diff_language(self):
        self.assertEqual(IsValidPassword("ಅಆಇಅಆಇಅ1Aa$%"),"The password is valid")

    def test_short_len_pwd(self):
        self.assertEqual(IsValidPassword("Shean123*"),['Length must be between 12 and
72 characters'])

    def test_long_len_pwd(self):
        self.assertEqual(IsValidPassword("Shean123*" + "a"*70),['Length must be between
12 and 72 characters'])

    def test_working_pwd(self):
        self.assertEqual(IsValidPassword("Shean12354*ab"),"The password is valid")

    def test_mix_languages(self):
        self.assertEqual(IsValidPassword("ಅಆಇಅ123loremipsumZZ*"),"The password is
valid")

    def test_only_symbols(self):
        self.assertEqual(IsValidPassword("!@#$%^&*()*!@"),['Password must contain a
lowercase letter','Password must contain an uppercase letter','Password must contain a
special character or a number'])

    def test_spaces_in_pwd(self):
        self.assertEqual(IsValidPassword("Shean 12354*ab"),"The password is valid")

if __name__ == "__main__":
    unittest.main()
```

Test suite: manual → Implementation: mistral

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors

    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")

    has_lower = any(c.islower() for c in password)
    if not has_lower:
        errors.append("Password must contain a lowercase letter")

    has_upper = any(c.isupper() for c in password)
    if not has_upper:
        errors.append("Password must contain an uppercase letter")

    has_special_or_digit = any(c.isdigit() or c in "!?@#$%^&*_-" for c in password)
    if not has_special_or_digit:
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"
    return errors
```

selected_test.py

```
# Tests for task6_validate_password
import unittest
from implementation_manual import IsValidPassword

class Test_password_valid(unittest.TestCase):
    def test_no_password(self):
        self.assertEqual(IsValidPassword(""),"Please fill the password")

    def test_no_lowercase(self):
        self.assertEqual(IsValidPassword("ABCDEFGHIJK*1"),['Password must contain a
lowercase letter'])

    def test_no_upprcase(self):
        self.assertEqual(IsValidPassword("abcdefghijk*1"),['Password must contain an
uppercase letter'])

    def test_no_symbols(self):
        self.assertEqual(IsValidPassword("Abcdefghijkl1"),['Password must contain a
special character or a number'])

    def test_lower_upper_no_symbol(self):
        self.assertEqual(IsValidPassword("Abcdefghijkl1"),['Password must contain a
```

```
special character or a number'])

    def test_nolower_upper_symbol(self):
        self.assertEqual(IsValidPassword("ABCDEFGHIJK*1"),['Password must contain a
lowercase letter'])

    def test_noupper_lower_symbol(self):
        self.assertEqual(IsValidPassword("abcdefghijk*1"),['Password must contain an
uppercase letter'])

    def test_diff_language(self):
        self.assertEqual(IsValidPassword("ಅಆಇಈಉಊಋ1Aa"),"The password is valid")

    def test_short_len_pwd(self):
        self.assertEqual(IsValidPassword("Shean123*"),['Length must be between 12 and
72 characters'])

    def test_long_len_pwd(self):
        self.assertEqual(IsValidPassword("Shean123*" + "a"*70),['Length must be between
12 and 72 characters'])

    def test_working_pwd(self):
        self.assertEqual(IsValidPassword("Shean12354*ab"),"The password is valid")

    def test_mix_languages(self):
        self.assertEqual(IsValidPassword("ಅಆಇಈ1231oremipsumZZ*"),"The password is
valid")

    def test_only_symbols(self):
        self.assertEqual(IsValidPassword("!@#$%^&*()*!@"),['Password must contain a
lowercase letter','Password must contain an uppercase letter','Password must contain a
special character or a number'])

    def test_spaces_in_pwd(self):
        self.assertEqual(IsValidPassword("Shean 12354*ab"),"The password is valid")

if __name__ == "__main__":
    unittest.main()
```

Test suite: manual → Implementation: openai

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors
    if not isinstance(password, str):
        return None
    length = len(password)
    if length < 12 or length > 72:
        errors.append("Length must be between 12 and 72 characters")
    if not any(c.islower() for c in password):
        errors.append("Password must contain a lowercase letter")
    if not any(c.isupper() for c in password):
        errors.append("Password must contain an uppercase letter")
    if not any(c.isdigit() or c in "!@#$%^&*_?" for c in password):
        errors.append("Password must contain a special character or a number")
    if errors:
        return errors
    return "The password is valid"
```

selected_test.py

```
# Tests for task6_validate_password
import unittest
from implementation_manual import IsValidPassword

class Test_password_valid(unittest.TestCase):
    def test_no_password(self):
        self.assertEqual(IsValidPassword(""),"Please fill the password")

    def test_no_lowercase(self):
        self.assertEqual(IsValidPassword("ABCDEFGHIJK*1"),['Password must contain a
lowercase letter'])

    def test_no_upprcase(self):
        self.assertEqual(IsValidPassword("abcdefghijk*1"),['Password must contain an
uppercase letter'])

    def test_no_symbols(self):
        self.assertEqual(IsValidPassword("Abcdefghijkl1"),['Password must contain a
special character or a number'])

    def test_lower_upper_no_symbol(self):
        self.assertEqual(IsValidPassword("Abcdefghijkl1"),['Password must contain a
special character or a number'])

    def test_nolower_upper_symbol(self):
        self.assertEqual(IsValidPassword("ABCDEFGHIJK*1"),['Password must contain a
lowercase letter'])
```

```
    def test_noupper_lower_symbol(self):
        self.assertEqual(IsValidPassword("abcdefghijk*1"),['Password must contain an
uppercase letter'])

    def test_diff_language(self):
        self.assertEqual(IsValidPassword("ಅಆಇಅಆಇಅಆಇ1Aa*"),"The password is valid")

    def test_short_len_pwd(self):
        self.assertEqual(IsValidPassword("Shean123*"),['Length must be between 12 and
72 characters'])

    def test_long_len_pwd(self):
        self.assertEqual(IsValidPassword("Shean123*" + "a"*70),['Length must be between
12 and 72 characters'])

    def test_working_pwd(self):
        self.assertEqual(IsValidPassword("Shean12354*ab"),"The password is valid")

    def test_mix_languages(self):
        self.assertEqual(IsValidPassword("ಅಆಇ1231oremipsumZZ*"),"The password is
valid")

    def test_only_symbols(self):
        self.assertEqual(IsValidPassword("!@#$%^&*()*!@"),['Password must contain a
lowercase letter','Password must contain an uppercase letter','Password must contain a
special character or a number'])

    def test_spaces_in_pwd(self):
        self.assertEqual(IsValidPassword("Shean 12354*ab"),"The password is valid")

if __name__ == "__main__":
    unittest.main()
```

Test suite: mistral → Implementation: amazon

implementation_manual.py

```
def is_valid_password(password):
    if password is None or password == "":
        return ["Please fill the password"]
    errors = []
    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")
    if not any(char.islower() for char in password):
        errors.append("Password must contain a lowercase letter")
    if not any(char.isupper() for char in password):
        errors.append("Password must contain an uppercase letter")
    if not any(char.isdigit() or not char.isalpha() for char in password):
        errors.append("Password must contain a special character or a number")
    if not errors:
        return "The password is valid"
    return errors
```

selected_test.py

```
from implementation_manual import IsValidPassword
import unittest

class TestIsValidPassword(unittest.TestCase):

    def test_missing_password(self):
        result = IsValidPassword("")
        self.assertEqual(result, ["Please fill the password"])

    def test_password_none(self):
        result = IsValidPassword(None)
        self.assertEqual(result, ["Please fill the password"])

    def test_password_too_short(self):
        result = IsValidPassword("short")
        self.assertEqual(result, ["Length must be between 12 and 72 characters"])

    def test_password_too_long(self):
        long_password = "a" * 73
        result = IsValidPassword(long_password)
        self.assertEqual(result, ["Length must be between 12 and 72 characters"])

    def test_password_exactly_12_chars(self):
        result = IsValidPassword("Aa1!Aa1!Aa1!A")
        self.assertEqual(result, "The password is valid")

    def test_password_exactly_72_chars(self):
        result =
IsValidPassword("Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!")
        self.assertEqual(result, "The password is valid")
```

```
    def test_missing_lowercase(self):
        result = IsValidPassword("ALLUPPERCASE123!")
        self.assertEqual(result, ["Password must contain a lowercase letter"])

    def test_missing_uppercase(self):
        result = IsValidPassword("alllowercase123!")
        self.assertEqual(result, ["Password must contain an uppercase letter"])

    def test_missing_special_or_number(self):
        result = IsValidPassword("AllUppercaseLowercase")
        self.assertEqual(result, ["Password must contain a special character or a
number"])

    def test_missing_multiple_requirements(self):
        result = IsValidPassword("short")
        self.assertEqual(result, [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ])

    def test_valid_password_with_all_requirements(self):
        result = IsValidPassword("ValidPass123!")
        self.assertEqual(result, "The password is valid")

    def test_valid_password_with_special_chars(self):
        result = IsValidPassword("Another@Valid1")
        self.assertEqual(result, "The password is valid")

    def test_password_with_spaces(self):
        result = IsValidPassword(" Valid Pass123! ")
        self.assertEqual(result, "The password is valid")

    def test_password_with_only_special_chars(self):
        result = IsValidPassword("!@#$%^&*()_+")
        self.assertEqual(result, [
            "Length must be between 12 and 72 characters",
            "Password must contain a lowercase letter",
            "Password must contain an uppercase letter"
        ])
```

Test suite: mistral → Implementation: deepseek

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors

    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")

    has_lower = any(c.islower() for c in password)
    has_upper = any(c.isupper() for c in password)
    has_digit_or_special = any(not c.isalpha() for c in password)

    if not has_lower:
        errors.append("Password must contain a lowercase letter")
    if not has_upper:
        errors.append("Password must contain an uppercase letter")
    if not has_digit_or_special:
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"
    else:
        return errors
```

selected_test.py

```
from implementation_manual import IsValidPassword
import unittest

class TestIsValidPassword(unittest.TestCase):

    def test_missing_password(self):
        result = IsValidPassword("")
        self.assertEqual(result, ["Please fill the password"])

    def test_password_none(self):
        result = IsValidPassword(None)
        self.assertEqual(result, ["Please fill the password"])

    def test_password_too_short(self):
        result = IsValidPassword("short")
        self.assertEqual(result, ["Length must be between 12 and 72 characters"])

    def test_password_too_long(self):
        long_password = "a" * 73
        result = IsValidPassword(long_password)
        self.assertEqual(result, ["Length must be between 12 and 72 characters"])
```

```
    def test_password_exactly_12_chars(self):
        result = IsValidPassword("Aa1!Aa1!Aa1!A")
        self.assertEqual(result, "The password is valid")

    def test_password_exactly_72_chars(self):
        result =
IsValidPassword("Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!")
        self.assertEqual(result, "The password is valid")

    def test_missing_lowercase(self):
        result = IsValidPassword("ALLUPPERCASE123!")
        self.assertEqual(result, ["Password must contain a lowercase letter"])

    def test_missing_uppercase(self):
        result = IsValidPassword("alllowercase123!")
        self.assertEqual(result, ["Password must contain an uppercase letter"])

    def test_missing_special_or_number(self):
        result = IsValidPassword("AllUppercaseLowercase")
        self.assertEqual(result, ["Password must contain a special character or a
number"])

    def test_missing_multiple_requirements(self):
        result = IsValidPassword("short")
        self.assertEqual(result, [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ])

    def test_valid_password_with_all_requirements(self):
        result = IsValidPassword("ValidPass123!")
        self.assertEqual(result, "The password is valid")

    def test_valid_password_with_special_chars(self):
        result = IsValidPassword("Another@Valid1")
        self.assertEqual(result, "The password is valid")

    def test_password_with_spaces(self):
        result = IsValidPassword(" Valid Pass123! ")
        self.assertEqual(result, "The password is valid")

    def test_password_with_only_special_chars(self):
        result = IsValidPassword("!@#$%^&*()_+")
        self.assertEqual(result, [
            "Length must be between 12 and 72 characters",
            "Password must contain a lowercase letter",
            "Password must contain an uppercase letter"
        ])
```

Test suite: mistral → Implementation: gemma

implementation_manual.py

```
import re

def IsValidPassword(password):
    if password is None or password == "":
        return ["Please fill the password"]

    errors = []

    if not (12 <= len(password) <= 72):
        errors.append("Length must be between 12 and 72 characters")

    if not re.search(r"[a-z]", password):
        errors.append("Password must contain a lowercase letter")

    if not re.search(r"[A-Z]", password):
        errors.append("Password must contain an uppercase letter")

    if not re.search(r"[!?@#$%^&*()_+0-9\-=_~`\[\]{}|;':\",./<>?]", password) and not
any(char.isdigit() for char in password):
        # The requirement specifies: one digit or punctuation character ([!? @#$ %^&* _
0 -9])
        # Let's use a regex that covers digits and the specified special characters
        if not re.search(r"[!?@#$%^&* _0-9]", password):
            errors.append("Password must contain a special character or a number")

    # Re-evaluating the special character/number requirement based on the prompt's
specific list
    # Prompt: "one digit or punctuation character ([!? @#$ %^&* _ 0 -9])"
    # Let's refine the check to be exact to the prompt's logic

    # Resetting error list for the specific character check to ensure accuracy
    # We need to check lowercase, uppercase, and (digit OR special)

    # Re-implementing logic cleanly
    errors = []

    if not (12 <= len(password) <= 72):
        errors.append("Length must be between 12 and 72 characters")

    if not any(c.islower() for c in password):
        errors.append("Password must contain a lowercase letter")

    if not any(c.isupper() for c in password):
        errors.append("Password must contain an uppercase letter")

    # Check for digit or the specific special characters provided
    special_chars = "!?@#$%^&*_"
    has_digit = any(c.isdigit() for c in password)
    has_special = any(c in special_chars for c in password)
```

```
    if not (has_digit or has_special):
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"

    return errors
```

selected_test.py

```
from implementation_manual import IsValidPassword
import unittest

class TestIsValidPassword(unittest.TestCase):

    def test_missing_password(self):
        result = IsValidPassword("")
        self.assertEqual(result, ["Please fill the password"])

    def test_password_none(self):
        result = IsValidPassword(None)
        self.assertEqual(result, ["Please fill the password"])

    def test_password_too_short(self):
        result = IsValidPassword("short")
        self.assertEqual(result, ["Length must be between 12 and 72 characters"])

    def test_password_too_long(self):
        long_password = "a" * 73
        result = IsValidPassword(long_password)
        self.assertEqual(result, ["Length must be between 12 and 72 characters"])

    def test_password_exactly_12_chars(self):
        result = IsValidPassword("Aa1!Aa1!Aa1!A")
        self.assertEqual(result, "The password is valid")

    def test_password_exactly_72_chars(self):
        result =
IsValidPassword("Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!")
        self.assertEqual(result, "The password is valid")

    def test_missing_lowercase(self):
        result = IsValidPassword("ALLUPPERCASE123!")
        self.assertEqual(result, ["Password must contain a lowercase letter"])

    def test_missing_uppercase(self):
        result = IsValidPassword("alllowercase123!")
        self.assertEqual(result, ["Password must contain an uppercase letter"])

    def test_missing_special_or_number(self):
        result = IsValidPassword("AllUppercaseLowercase")
        self.assertEqual(result, ["Password must contain a special character or a
```

```
number"])

    def test_missing_multiple_requirements(self):
        result = IsValidPassword("short")
        self.assertEqual(result, [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ])

    def test_valid_password_with_all_requirements(self):
        result = IsValidPassword("ValidPass123!")
        self.assertEqual(result, "The password is valid")

    def test_valid_password_with_special_chars(self):
        result = IsValidPassword("Another@Valid1")
        self.assertEqual(result, "The password is valid")

    def test_password_with_spaces(self):
        result = IsValidPassword(" Valid Pass123! ")
        self.assertEqual(result, "The password is valid")

    def test_password_with_only_special_chars(self):
        result = IsValidPassword("!@#$%^&*()_+")
        self.assertEqual(result, [
            "Length must be between 12 and 72 characters",
            "Password must contain a lowercase letter",
            "Password must contain an uppercase letter"
        ])
```

Test suite: mistral → Implementation: manual

implementation_manual.py

```
def IsValidPassword(password):
    errors=[]
    if len(password)==0:
        return "Please fill the password"
    if len(password)<12 or len(password)>72:
        errors.append("Length must be between 12 and 72 characters")
    lowerCase=False
    upperCase=False
    num=False
    special=False
    for char in password:
        if(char.isdigit()):
            num=True
        if(char.islower()):
            lowerCase=True
        if(char.isupper()):
            upperCase=True
        if(not char.isalnum()):
            special=True
    if not lowerCase:
        errors.append("Password must contain a lowercase letter")
    if not upperCase:
        errors.append("Password must contain an uppercase letter")
    if not special or not num:
        errors.append("Password must contain a special character or a number")
    if len(errors)==0:
        return "The password is valid"
    else:
        return errors
```

selected_test.py

```
from implementation_manual import IsValidPassword
import unittest

class TestIsValidPassword(unittest.TestCase):

    def test_missing_password(self):
        result = IsValidPassword("")
        self.assertEqual(result, ["Please fill the password"])

    def test_password_none(self):
        result = IsValidPassword(None)
        self.assertEqual(result, ["Please fill the password"])

    def test_password_too_short(self):
        result = IsValidPassword("short")
```

```
        self.assertEqual(result, ["Length must be between 12 and 72 characters"])

    def test_password_too_long(self):
        long_password = "a" * 73
        result = IsValidPassword(long_password)
        self.assertEqual(result, ["Length must be between 12 and 72 characters"])

    def test_password_exactly_12_chars(self):
        result = IsValidPassword("Aa1!Aa1!Aa1!A")
        self.assertEqual(result, "The password is valid")

    def test_password_exactly_72_chars(self):
        result =
IsValidPassword("Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!")
        self.assertEqual(result, "The password is valid")

    def test_missing_lowercase(self):
        result = IsValidPassword("ALLUPPERCASE123!")
        self.assertEqual(result, ["Password must contain a lowercase letter"])

    def test_missing_uppercase(self):
        result = IsValidPassword("alllowercase123!")
        self.assertEqual(result, ["Password must contain an uppercase letter"])

    def test_missing_special_or_number(self):
        result = IsValidPassword("AllUppercaseLowercase")
        self.assertEqual(result, ["Password must contain a special character or a
number"])

    def test_missing_multiple_requirements(self):
        result = IsValidPassword("short")
        self.assertEqual(result, [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ])

    def test_valid_password_with_all_requirements(self):
        result = IsValidPassword("ValidPass123!")
        self.assertEqual(result, "The password is valid")

    def test_valid_password_with_special_chars(self):
        result = IsValidPassword("Another@Valid1")
        self.assertEqual(result, "The password is valid")

    def test_password_with_spaces(self):
        result = IsValidPassword(" Valid Pass123! ")
        self.assertEqual(result, "The password is valid")

    def test_password_with_only_special_chars(self):
        result = IsValidPassword("!@#$%^&*()_+")
        self.assertEqual(result, [
            "Length must be between 12 and 72 characters",
```

```
    "Password must contain a lowercase letter",
    "Password must contain an uppercase letter"
])
```

Test suite: mistral → Implementation: mistral

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors

    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")

    has_lower = any(c.islower() for c in password)
    if not has_lower:
        errors.append("Password must contain a lowercase letter")

    has_upper = any(c.isupper() for c in password)
    if not has_upper:
        errors.append("Password must contain an uppercase letter")

    has_special_or_digit = any(c.isdigit() or c in "!?@#$%^&*_-" for c in password)
    if not has_special_or_digit:
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"
    return errors
```

selected_test.py

```
from implementation_manual import IsValidPassword
import unittest

class TestIsValidPassword(unittest.TestCase):

    def test_missing_password(self):
        result = IsValidPassword("")
        self.assertEqual(result, ["Please fill the password"])

    def test_password_none(self):
        result = IsValidPassword(None)
        self.assertEqual(result, ["Please fill the password"])

    def test_password_too_short(self):
        result = IsValidPassword("short")
        self.assertEqual(result, ["Length must be between 12 and 72 characters"])

    def test_password_too_long(self):
        long_password = "a" * 73
        result = IsValidPassword(long_password)
        self.assertEqual(result, ["Length must be between 12 and 72 characters"])
```

```
    def test_password_exactly_12_chars(self):
        result = IsValidPassword("Aa1!Aa1!Aa1!A")
        self.assertEqual(result, "The password is valid")

    def test_password_exactly_72_chars(self):
        result =
IsValidPassword("Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!")
        self.assertEqual(result, "The password is valid")

    def test_missing_lowercase(self):
        result = IsValidPassword("ALLUPPERCASE123!")
        self.assertEqual(result, ["Password must contain a lowercase letter"])

    def test_missing_uppercase(self):
        result = IsValidPassword("alllowercase123!")
        self.assertEqual(result, ["Password must contain an uppercase letter"])

    def test_missing_special_or_number(self):
        result = IsValidPassword("AllUppercaseLowercase")
        self.assertEqual(result, ["Password must contain a special character or a
number"])

    def test_missing_multiple_requirements(self):
        result = IsValidPassword("short")
        self.assertEqual(result, [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ])

    def test_valid_password_with_all_requirements(self):
        result = IsValidPassword("ValidPass123!")
        self.assertEqual(result, "The password is valid")

    def test_valid_password_with_special_chars(self):
        result = IsValidPassword("Another@Valid1")
        self.assertEqual(result, "The password is valid")

    def test_password_with_spaces(self):
        result = IsValidPassword(" Valid Pass123! ")
        self.assertEqual(result, "The password is valid")

    def test_password_with_only_special_chars(self):
        result = IsValidPassword("!@#$%^&*()_+")
        self.assertEqual(result, [
            "Length must be between 12 and 72 characters",
            "Password must contain a lowercase letter",
            "Password must contain an uppercase letter"
        ])
```

Test suite: mistral → Implementation: openai

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors
    if not isinstance(password, str):
        return None
    length = len(password)
    if length < 12 or length > 72:
        errors.append("Length must be between 12 and 72 characters")
    if not any(c.islower() for c in password):
        errors.append("Password must contain a lowercase letter")
    if not any(c.isupper() for c in password):
        errors.append("Password must contain an uppercase letter")
    if not any(c.isdigit() or c in "!@#$%^&*_?" for c in password):
        errors.append("Password must contain a special character or a number")
    if errors:
        return errors
    return "The password is valid"
```

selected_test.py

```
from implementation_manual import IsValidPassword
import unittest

class TestIsValidPassword(unittest.TestCase):

    def test_missing_password(self):
        result = IsValidPassword("")
        self.assertEqual(result, ["Please fill the password"])

    def test_password_none(self):
        result = IsValidPassword(None)
        self.assertEqual(result, ["Please fill the password"])

    def test_password_too_short(self):
        result = IsValidPassword("short")
        self.assertEqual(result, ["Length must be between 12 and 72 characters"])

    def test_password_too_long(self):
        long_password = "a" * 73
        result = IsValidPassword(long_password)
        self.assertEqual(result, ["Length must be between 12 and 72 characters"])

    def test_password_exactly_12_chars(self):
        result = IsValidPassword("Aa1!Aa1!Aa1!A")
        self.assertEqual(result, "The password is valid")

    def test_password_exactly_72_chars(self):
```

```
        result =
IsValidPassword("Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!Aa1!")
        self.assertEqual(result, "The password is valid")

    def test_missing_lowercase(self):
        result = IsValidPassword("ALLUPPERCASE123!")
        self.assertEqual(result, ["Password must contain a lowercase letter"])

    def test_missing_uppercase(self):
        result = IsValidPassword("alllowercase123!")
        self.assertEqual(result, ["Password must contain an uppercase letter"])

    def test_missing_special_or_number(self):
        result = IsValidPassword("AllUppercaseLowercase")
        self.assertEqual(result, ["Password must contain a special character or a
number"])

    def test_missing_multiple_requirements(self):
        result = IsValidPassword("short")
        self.assertEqual(result, [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ])

    def test_valid_password_with_all_requirements(self):
        result = IsValidPassword("ValidPass123!")
        self.assertEqual(result, "The password is valid")

    def test_valid_password_with_special_chars(self):
        result = IsValidPassword("Another@Valid1")
        self.assertEqual(result, "The password is valid")

    def test_password_with_spaces(self):
        result = IsValidPassword(" Valid Pass123! ")
        self.assertEqual(result, "The password is valid")

    def test_password_with_only_special_chars(self):
        result = IsValidPassword("!@#$%^&*()_+")
        self.assertEqual(result, [
            "Length must be between 12 and 72 characters",
            "Password must contain a lowercase letter",
            "Password must contain an uppercase letter"
        ])
```

Test suite: openai → Implementation: amazon

implementation_manual.py

```
def is_valid_password(password):
    if password is None or password == "":
        return ["Please fill the password"]
    errors = []
    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")
    if not any(char.islower() for char in password):
        errors.append("Password must contain a lowercase letter")
    if not any(char.isupper() for char in password):
        errors.append("Password must contain an uppercase letter")
    if not any(char.isdigit() or not char.isalpha() for char in password):
        errors.append("Password must contain a special character or a number")
    if not errors:
        return "The password is valid"
    return errors
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestIsValidPassword(unittest.TestCase):

    def test_missing_password(self):
        # Arrange
        password = ""
        expected_errors = ["Please fill the password"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_errors)

    def test_password_too_short(self):
        # Arrange
        password = "Ab1!"
        expected_errors = [
            "Length must be between 12 and 72 characters",
            "Password must contain a special character or a number"
        ]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Length must be between 12 and 72 characters", result)
        self.assertIn("Password must contain a special character or a number", result)
```

```
        self.assertIn("Password must contain a lowercase letter", result) or
self.assertIn("Password must contain an uppercase letter", result) or True

    def test_password_too_long(self):
        # Arrange
        password = "A" * 73 + "a1!"
        expected_errors = ["Length must be between 12 and 72 characters"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Length must be between 12 and 72 characters", result)

    def test_password_missing_lowercase(self):
        # Arrange
        password = "ABCDEFGHIJKL1!"
        expected_errors = ["Password must contain a lowercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Password must contain a lowercase letter", result)

    def test_password_missing_uppercase(self):
        # Arrange
        password = "abcdefghijk1!"
        expected_errors = ["Password must contain an uppercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Password must contain an uppercase letter", result)

    def test_password_missing_special_or_number(self):
        # Arrange
        password = "Abcdefghijkl"
        expected_errors = ["Password must contain a special character or a number"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Password must contain a special character or a number", result)

    def test_password_multiple_errors(self):
        # Arrange
        password = "abc"
        expected_errors = [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
```

```
            "Password must contain a special character or a number"
        ]

        # Act
        result = IsValidPassword(password)

        # Assert
        for error in expected_errors:
            self.assertIn(error, result)

    def test_valid_password(self):
        # Arrange
        password = "ValidPass123!"
        expected_message = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

    def test_password_at_minimum_length(self):
        # Arrange
        password = "Aabcde12345!"
        expected_message = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

    def test_password_at_maximum_length(self):
        # Arrange
        password = "A" + "a" * 69 + "1!"
        expected_message = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

if __name__ == "__main__":
    unittest.main()
```

Test suite: openai → Implementation: deepseek

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors

    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")

    has_lower = any(c.islower() for c in password)
    has_upper = any(c.isupper() for c in password)
    has_digit_or_special = any(not c.isalpha() for c in password)

    if not has_lower:
        errors.append("Password must contain a lowercase letter")
    if not has_upper:
        errors.append("Password must contain an uppercase letter")
    if not has_digit_or_special:
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"
    else:
        return errors
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestIsValidPassword(unittest.TestCase):

    def test_missing_password(self):
        # Arrange
        password = ""
        expected_errors = ["Please fill the password"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_errors)

    def test_password_too_short(self):
        # Arrange
        password = "Ab1!"
        expected_errors = [
            "Length must be between 12 and 72 characters",
            "Password must contain a special character or a number"
```

```
        ]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Length must be between 12 and 72 characters", result)
        self.assertIn("Password must contain a special character or a number", result)
        self.assertIn("Password must contain a lowercase letter", result) or
self.assertIn("Password must contain an uppercase letter", result) or True

    def test_password_too_long(self):
        # Arrange
        password = "A" * 73 + "a1!"
        expected_errors = ["Length must be between 12 and 72 characters"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Length must be between 12 and 72 characters", result)

    def test_password_missing_lowercase(self):
        # Arrange
        password = "ABCDEFGHIJKL1!"
        expected_errors = ["Password must contain a lowercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Password must contain a lowercase letter", result)

    def test_password_missing_uppercase(self):
        # Arrange
        password = "abcdefghijk1!"
        expected_errors = ["Password must contain an uppercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Password must contain an uppercase letter", result)

    def test_password_missing_special_or_number(self):
        # Arrange
        password = "Abcdefghijkl"
        expected_errors = ["Password must contain a special character or a number"]

        # Act
        result = IsValidPassword(password)

        # Assert
```

```
        self.assertIn("Password must contain a special character or a number", result)

    def test_password_multiple_errors(self):
        # Arrange
        password = "abc"
        expected_errors = [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ]

        # Act
        result = IsValidPassword(password)

        # Assert
        for error in expected_errors:
            self.assertIn(error, result)

    def test_valid_password(self):
        # Arrange
        password = "ValidPass123!"
        expected_message = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

    def test_password_at_minimum_length(self):
        # Arrange
        password = "Aabcde12345!"
        expected_message = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

    def test_password_at_maximum_length(self):
        # Arrange
        password = "A" + "a" * 69 + "1!"
        expected_message = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

if __name__ == "__main__":
    unittest.main()
```

Test suite: openai → Implementation: gemma

implementation_manual.py

```
import re

def IsValidPassword(password):
    if password is None or password == "":
        return ["Please fill the password"]

    errors = []

    if not (12 <= len(password) <= 72):
        errors.append("Length must be between 12 and 72 characters")

    if not re.search(r"[a-z]", password):
        errors.append("Password must contain a lowercase letter")

    if not re.search(r"[A-Z]", password):
        errors.append("Password must contain an uppercase letter")

    if not re.search(r"[!?@#$%^&*()_+0-9\-=_~`\[\]{}|;':\",./<>?]", password) and not
any(char.isdigit() for char in password):
        # The requirement specifies: one digit or punctuation character ([!? @#$ %^&* _
0 -9])
        # Let's use a regex that covers digits and the specified special characters
        if not re.search(r"[!?@#$%^&* _0-9]", password):
            errors.append("Password must contain a special character or a number")

    # Re-evaluating the special character/number requirement based on the prompt's
specific list
    # Prompt: "one digit or punctuation character ([!? @#$ %^&* _ 0 -9])"
    # Let's refine the check to be exact to the prompt's logic

    # Resetting error list for the specific character check to ensure accuracy
    # We need to check lowercase, uppercase, and (digit OR special)

    # Re-implementing logic cleanly
    errors = []

    if not (12 <= len(password) <= 72):
        errors.append("Length must be between 12 and 72 characters")

    if not any(c.islower() for c in password):
        errors.append("Password must contain a lowercase letter")

    if not any(c.isupper() for c in password):
        errors.append("Password must contain an uppercase letter")

    # Check for digit or the specific special characters provided
    special_chars = "!?@#$%^&*_"
    has_digit = any(c.isdigit() for c in password)
    has_special = any(c in special_chars for c in password)
```

```
    if not (has_digit or has_special):
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"

    return errors
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestIsValidPassword(unittest.TestCase):

    def test_missing_password(self):
        # Arrange
        password = ""
        expected_errors = ["Please fill the password"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_errors)

    def test_password_too_short(self):
        # Arrange
        password = "Ab1!"
        expected_errors = [
            "Length must be between 12 and 72 characters",
            "Password must contain a special character or a number"
        ]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Length must be between 12 and 72 characters", result)
        self.assertIn("Password must contain a special character or a number", result)
        self.assertIn("Password must contain a lowercase letter", result) or
self.assertIn("Password must contain an uppercase letter", result) or True

    def test_password_too_long(self):
        # Arrange
        password = "A" * 73 + "a1!"
        expected_errors = ["Length must be between 12 and 72 characters"]

        # Act
        result = IsValidPassword(password)

        # Assert
```

```
        self.assertIn("Length must be between 12 and 72 characters", result)

    def test_password_missing_lowercase(self):
        # Arrange
        password = "ABCDEFGHIJKL1!"
        expected_errors = ["Password must contain a lowercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Password must contain a lowercase letter", result)

    def test_password_missing_uppercase(self):
        # Arrange
        password = "abcdefghijk1!"
        expected_errors = ["Password must contain an uppercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Password must contain an uppercase letter", result)

    def test_password_missing_special_or_number(self):
        # Arrange
        password = "Abcdefghijkl"
        expected_errors = ["Password must contain a special character or a number"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Password must contain a special character or a number", result)

    def test_password_multiple_errors(self):
        # Arrange
        password = "abc"
        expected_errors = [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ]

        # Act
        result = IsValidPassword(password)

        # Assert
        for error in expected_errors:
            self.assertIn(error, result)

    def test_valid_password(self):
        # Arrange
```

```
        password = "ValidPass123!"
        expected_message = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

    def test_password_at_minimum_length(self):
        # Arrange
        password = "Aabcde12345!"
        expected_message = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

    def test_password_at_maximum_length(self):
        # Arrange
        password = "A" + "a" * 69 + "1!"
        expected_message = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

if __name__ == "__main__":
    unittest.main()
```

Test suite: openai → Implementation: manual

implementation_manual.py

```
def IsValidPassword(password):
    errors=[]
    if len(password)==0:
        return "Please fill the password"
    if len(password)<12 or len(password)>72:
        errors.append("Length must be between 12 and 72 characters")
    lowerCase=False
    upperCase=False
    num=False
    special=False
    for char in password:
        if(char.isdigit()):
            num=True
        if(char.islower()):
            lowerCase=True
        if(char.isupper()):
            upperCase=True
        if(not char.isalnum()):
            special=True
    if not lowerCase:
        errors.append("Password must contain a lowercase letter")
    if not upperCase:
        errors.append("Password must contain an uppercase letter")
    if not special or not num:
        errors.append("Password must contain a special character or a number")
    if len(errors)==0:
        return "The password is valid"
    else:
        return errors
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestIsValidPassword(unittest.TestCase):

    def test_missing_password(self):
        # Arrange
        password = ""
        expected_errors = ["Please fill the password"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_errors)
```

```
    def test_password_too_short(self):
        # Arrange
        password = "Ab1!"
        expected_errors = [
            "Length must be between 12 and 72 characters",
            "Password must contain a special character or a number"
        ]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Length must be between 12 and 72 characters", result)
        self.assertIn("Password must contain a special character or a number", result)
        self.assertIn("Password must contain a lowercase letter", result) or
self.assertIn("Password must contain an uppercase letter", result) or True

    def test_password_too_long(self):
        # Arrange
        password = "A" * 73 + "a1!"
        expected_errors = ["Length must be between 12 and 72 characters"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Length must be between 12 and 72 characters", result)

    def test_password_missing_lowercase(self):
        # Arrange
        password = "ABCDEFGHIJKL1!"
        expected_errors = ["Password must contain a lowercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Password must contain a lowercase letter", result)

    def test_password_missing_uppercase(self):
        # Arrange
        password = "abcdefghijk1!"
        expected_errors = ["Password must contain an uppercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Password must contain an uppercase letter", result)

    def test_password_missing_special_or_number(self):
        # Arrange
```

```
        password = "Abcdefghijkl"
        expected_errors = ["Password must contain a special character or a number"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Password must contain a special character or a number", result)

    def test_password_multiple_errors(self):
        # Arrange
        password = "abc"
        expected_errors = [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ]

        # Act
        result = IsValidPassword(password)

        # Assert
        for error in expected_errors:
            self.assertIn(error, result)

    def test_valid_password(self):
        # Arrange
        password = "ValidPass123!"
        expected_message = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

    def test_password_at_minimum_length(self):
        # Arrange
        password = "Aabcde12345!"
        expected_message = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

    def test_password_at_maximum_length(self):
        # Arrange
        password = "A" + "a" * 69 + "1!"
        expected_message = "The password is valid"

        # Act
```

```
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

if __name__ == "__main__":
    unittest.main()
```

Test suite: openai → Implementation: mistral

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors

    if len(password) < 12 or len(password) > 72:
        errors.append("Length must be between 12 and 72 characters")

    has_lower = any(c.islower() for c in password)
    if not has_lower:
        errors.append("Password must contain a lowercase letter")

    has_upper = any(c.isupper() for c in password)
    if not has_upper:
        errors.append("Password must contain an uppercase letter")

    has_special_or_digit = any(c.isdigit() or c in "!?@#$%^&*_-" for c in password)
    if not has_special_or_digit:
        errors.append("Password must contain a special character or a number")

    if not errors:
        return "The password is valid"
    return errors
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestIsValidPassword(unittest.TestCase):

    def test_missing_password(self):
        # Arrange
        password = ""
        expected_errors = ["Please fill the password"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_errors)

    def test_password_too_short(self):
        # Arrange
        password = "Ab1!"
        expected_errors = [
            "Length must be between 12 and 72 characters",
            "Password must contain a special character or a number"
```

```
        ]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Length must be between 12 and 72 characters", result)
        self.assertIn("Password must contain a special character or a number", result)
        self.assertIn("Password must contain a lowercase letter", result) or
self.assertIn("Password must contain an uppercase letter", result) or True

    def test_password_too_long(self):
        # Arrange
        password = "A" * 73 + "a1!"
        expected_errors = ["Length must be between 12 and 72 characters"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Length must be between 12 and 72 characters", result)

    def test_password_missing_lowercase(self):
        # Arrange
        password = "ABCDEFGHIJKL1!"
        expected_errors = ["Password must contain a lowercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Password must contain a lowercase letter", result)

    def test_password_missing_uppercase(self):
        # Arrange
        password = "abcdefghijk1!"
        expected_errors = ["Password must contain an uppercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Password must contain an uppercase letter", result)

    def test_password_missing_special_or_number(self):
        # Arrange
        password = "Abcdefghijkl"
        expected_errors = ["Password must contain a special character or a number"]

        # Act
        result = IsValidPassword(password)

        # Assert
```

```
        self.assertIn("Password must contain a special character or a number", result)

    def test_password_multiple_errors(self):
        # Arrange
        password = "abc"
        expected_errors = [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ]

        # Act
        result = IsValidPassword(password)

        # Assert
        for error in expected_errors:
            self.assertIn(error, result)

    def test_valid_password(self):
        # Arrange
        password = "ValidPass123!"
        expected_message = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

    def test_password_at_minimum_length(self):
        # Arrange
        password = "Aabcde12345!"
        expected_message = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

    def test_password_at_maximum_length(self):
        # Arrange
        password = "A" + "a" * 69 + "1!"
        expected_message = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

if __name__ == "__main__":
    unittest.main()
```

Test suite: openai → Implementation: openai

implementation_manual.py

```
def IsValidPassword(password):
    errors = []
    if not password:
        errors.append("Please fill the password")
        return errors
    if not isinstance(password, str):
        return None
    length = len(password)
    if length < 12 or length > 72:
        errors.append("Length must be between 12 and 72 characters")
    if not any(c.islower() for c in password):
        errors.append("Password must contain a lowercase letter")
    if not any(c.isupper() for c in password):
        errors.append("Password must contain an uppercase letter")
    if not any(c.isdigit() or c in "!@#$%^&*_?" for c in password):
        errors.append("Password must contain a special character or a number")
    if errors:
        return errors
    return "The password is valid"
```

selected_test.py

```
import unittest
from implementation_manual import IsValidPassword

class TestIsValidPassword(unittest.TestCase):

    def test_missing_password(self):
        # Arrange
        password = ""
        expected_errors = ["Please fill the password"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_errors)

    def test_password_too_short(self):
        # Arrange
        password = "Ab1!"
        expected_errors = [
            "Length must be between 12 and 72 characters",
            "Password must contain a special character or a number"
        ]

        # Act
        result = IsValidPassword(password)
```

```
        # Assert
        self.assertIn("Length must be between 12 and 72 characters", result)
        self.assertIn("Password must contain a special character or a number", result)
        self.assertIn("Password must contain a lowercase letter", result) or
self.assertIn("Password must contain an uppercase letter", result) or True

    def test_password_too_long(self):
        # Arrange
        password = "A" * 73 + "a1!"
        expected_errors = ["Length must be between 12 and 72 characters"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Length must be between 12 and 72 characters", result)

    def test_password_missing_lowercase(self):
        # Arrange
        password = "ABCDEFGHIJKL1!"
        expected_errors = ["Password must contain a lowercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Password must contain a lowercase letter", result)

    def test_password_missing_uppercase(self):
        # Arrange
        password = "abcdefghijk1!"
        expected_errors = ["Password must contain an uppercase letter"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Password must contain an uppercase letter", result)

    def test_password_missing_special_or_number(self):
        # Arrange
        password = "Abcdefghijkl"
        expected_errors = ["Password must contain a special character or a number"]

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertIn("Password must contain a special character or a number", result)

    def test_password_multiple_errors(self):
        # Arrange
        password = "abc"
```

```
        expected_errors = [
            "Length must be between 12 and 72 characters",
            "Password must contain an uppercase letter",
            "Password must contain a special character or a number"
        ]

        # Act
        result = IsValidPassword(password)

        # Assert
        for error in expected_errors:
            self.assertIn(error, result)

    def test_valid_password(self):
        # Arrange
        password = "ValidPass123!"
        expected_message = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

    def test_password_at_minimum_length(self):
        # Arrange
        password = "Aabcde12345!"
        expected_message = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

    def test_password_at_maximum_length(self):
        # Arrange
        password = "A" + "a" * 69 + "1!"
        expected_message = "The password is valid"

        # Act
        result = IsValidPassword(password)

        # Assert
        self.assertEqual(result, expected_message)

if __name__ == "__main__":
    unittest.main()
```